



Taller de Docker

Entregables:

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2. Bases de datos del proyecto en docker
3. Backend y Frontend IIS apuntando a la base de datos Docker
4. Backend y Frontend en Docker apuntando a la base de datos Docker

1. Motores de Bases de Datos en Docker

Vista general de contenedores en Docker Desktop: En la captura se ve el panel Containers de Docker Desktop con los motores de base de datos ejecutándose en contenedores separados. Esta vista confirma que los contenedores están arriba y listos para aceptar conexiones externas desde herramientas cliente o desde aplicaciones.

The screenshot shows the Docker Desktop interface. The left sidebar contains navigation options: Ask Gordon, Containers, Images, Volumes, Builds, Models, MCP Toolkit, Docker Hub, Docker Scout, and Extensions. The main panel is titled 'Containers' and displays a table of running containers. Above the table, it shows 'Container CPU usage' at 1.27% / 1200% (12 CPUs available) and 'Container memory usage' at 505.24MB / 7.4GB. A search bar and a toggle for 'Only show running containers' are also present. The table lists five containers: 'determined_noyce' (postgres:latest), 'motoresdocker' (postgres:latest), 'mysql_clean' (mysql:latest), 'sqlserver_clean' (mssql/server:2022), and 'postgres_clean' (postgres:latest). Each container entry includes its name, ID, image, ports, CPU usage, last started time, and actions (restart, stop, delete, etc.). At the bottom, there are 'Walkthroughs' for 'Multi-container applications' and 'Containerize your application'. The status bar at the very bottom indicates 'Engine running', system resources (RAM 5.69 GB, CPU 2.93%, Disk 5.76 GB used), and the date/time (02/09/2025, 18:21).

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	determined_noyce	dd287662e82a	postgres:latest		0%	8 minutes ago	
☒	motoresdocker	-	-	-	1.27%	1 minute ago	
☐	mysql_clean	0a9696c9d5bd	mysql:latest	3308:3306	1.27%	24 minutes ago	
☐	sqlserver_clean	490fa10954fb	mssql/server:2022	http://localhost:5434	0%	1 minute ago	
☐	postgres_clean	9c711ed7fa23	postgres:latest	5434:5432	0%	24 minutes ago	

Conexión a PostgreSQL

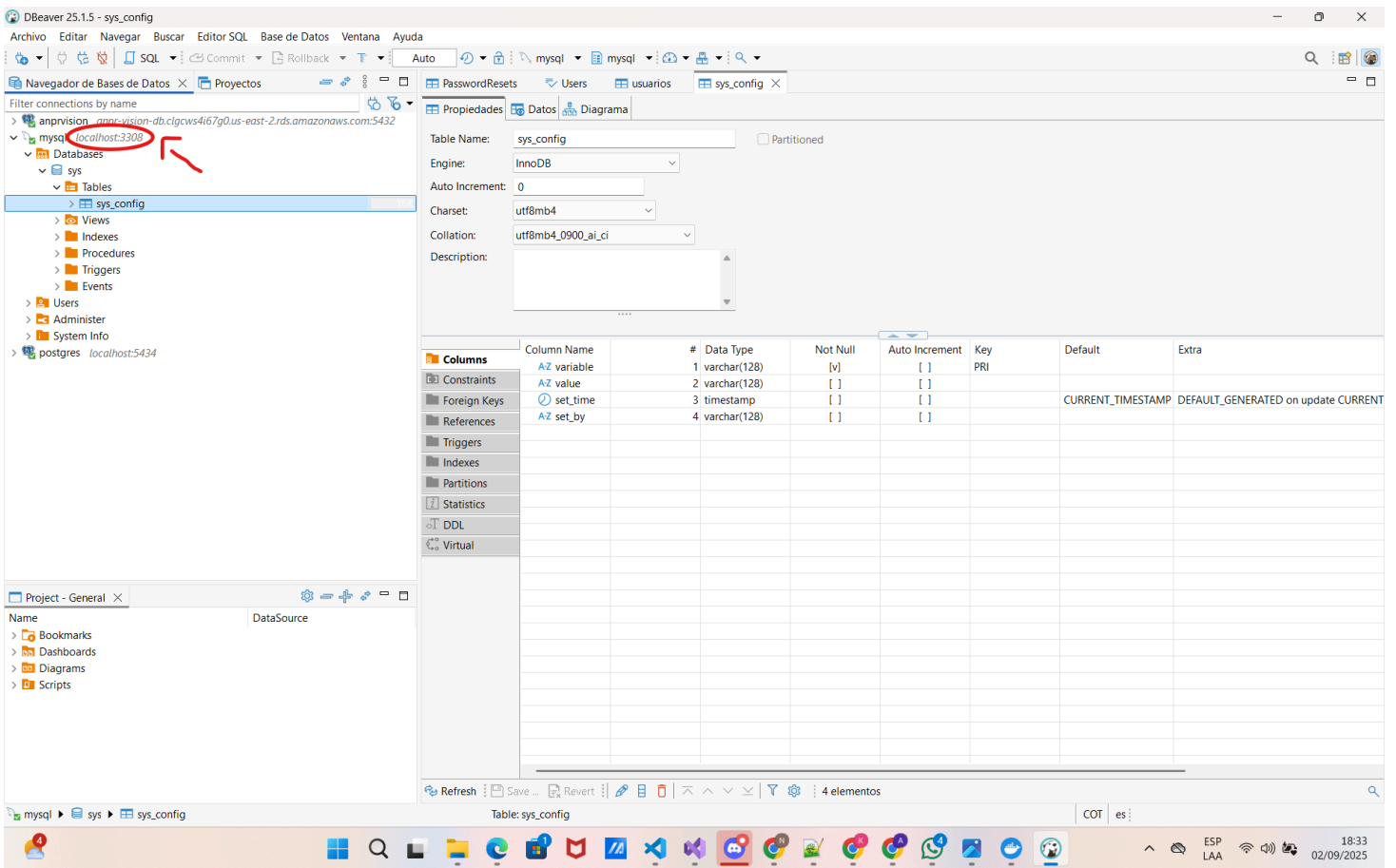
La parte superior muestra los logs del contenedor de PostgreSQL indicando que el servidor se levantó correctamente y quedó a la espera de conexiones. Abajo se ve la conexión configurada en el cliente (DBeaver/pgAdmin) usando Host localhost, Puerto 5434, y las credenciales definidas en el contenedor. Esta combinación verifica que el mapeo de puertos es el correcto y que el motor en Docker responde como si fuera una instalación local, permitiendo explorar esquemas, tablas y ejecutar consultas sin salir del entorno de contenedores.

The screenshot shows the Docker Desktop interface. On the left is a sidebar with navigation options: Ask Gordon, Containers, Images, Volumes, Builds, Models, MCP Toolkit, Docker Hub, Docker Scout, and Extensions. The main area displays the 'postgres_clean' container, which is running. The 'Logs' tab is selected, showing a detailed log of the PostgreSQL server's startup and operation. The logs indicate that the server started successfully, listening on port 5432 and the Unix socket. A red circle highlights the port mapping '5434:5432' in the container's configuration. The bottom status bar shows system metrics: RAM 5.69 GB, CPU 0.42%, and Disk 5.76 GB used. The system clock shows 18:22 on 02/09/2025.

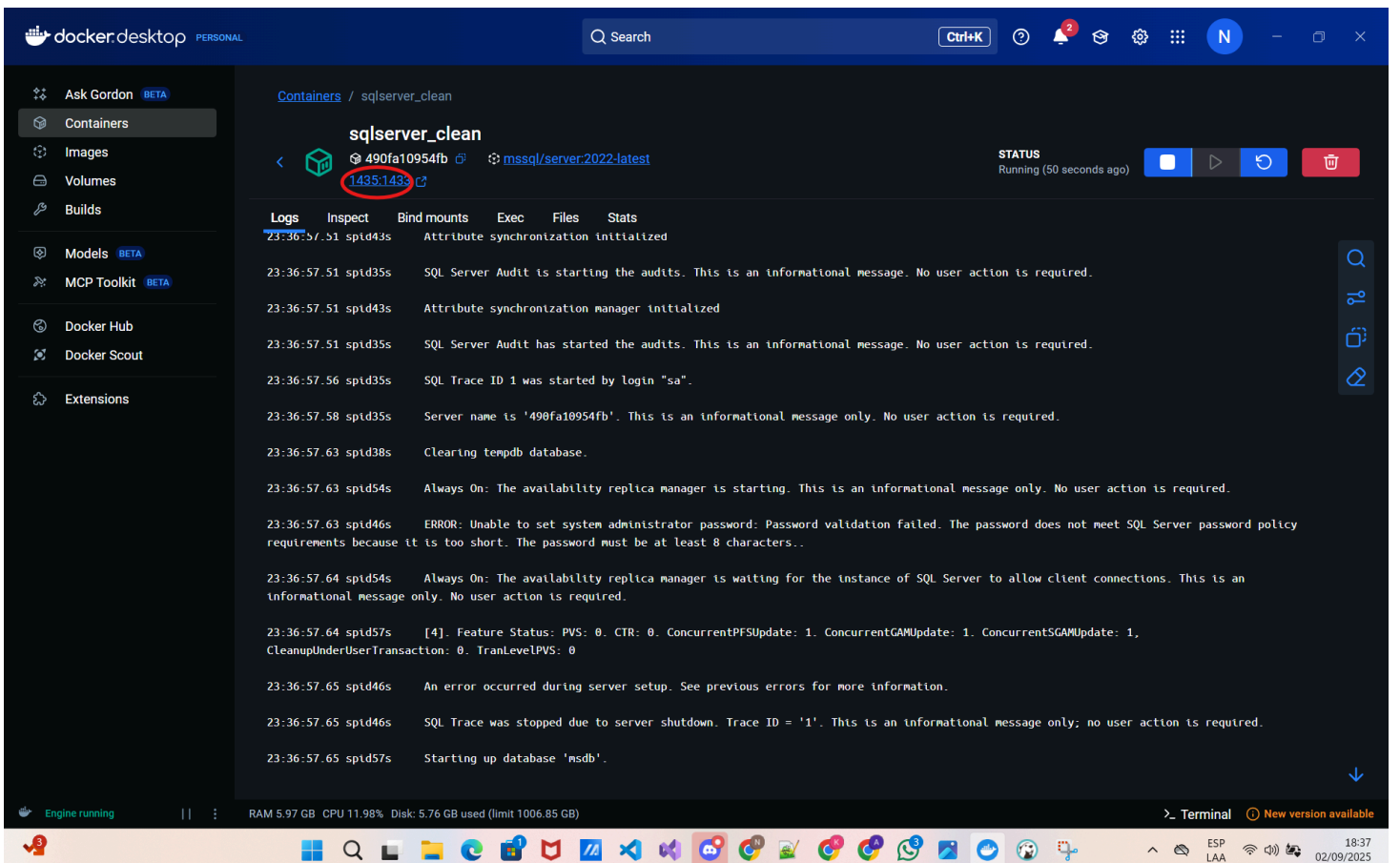
```
2025-08-25 03:09:21.861 UTC [63] LOG:  checkpoint complete: wrote 931 buffers (5.7%); 1 WAL file(s) added, 0 removed, 0 recycled; write=78.637 s, sync=0.438 s, total=79.174 s; sync files=311, longest=0.010 s, average=0.002 s; distance=4304 kB, estimate=4304 kB; lsn=0/1DA21E8, redo lsn=0/1D7C1F8
2025-08-25 03:11:55.187 UTC [111] FATAL:  role "postgres" does not exist
2025-08-25 03:12:48.444 UTC [120] FATAL:  role "motoredocker" does not exist
2025-08-25 03:40:49.310 UTC [1] LOG:  received fast shutdown request
2025-08-25 03:40:49.314 UTC [1] LOG:  aborting any active transactions
2025-08-25 03:40:49.322 UTC [95] FATAL:  terminating connection due to administrator command
2025-08-25 03:40:49.322 UTC [93] FATAL:  terminating connection due to administrator command
2025-08-25 03:40:49.322 UTC [90] FATAL:  terminating connection due to administrator command
2025-08-25 03:40:49.322 UTC [89] FATAL:  terminating connection due to administrator command
2025-08-25 03:40:49.323 UTC [92] FATAL:  terminating connection due to administrator command
2025-08-25 03:40:49.334 UTC [1] LOG:  background worker "logical replication launcher" (PID 68) exited with exit code 1
2025-08-25 03:40:49.342 UTC [63] LOG:  shutting down
2025-08-25 03:40:49.346 UTC [63] LOG:  checkpoint starting: shutdown immediate
2025-08-25 03:40:49.421 UTC [63] LOG:  checkpoint complete: wrote 48 buffers (0.3%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.014 s, sync=0.041 s, total=0.078 s; sync files=37, longest=0.010 s, average=0.002 s; distance=152 kB, estimate=3889 kB; lsn=0/1DA2298, redo lsn=0/1DA2298
2025-08-25 03:40:49.431 UTC [1] LOG:  database system is shut down

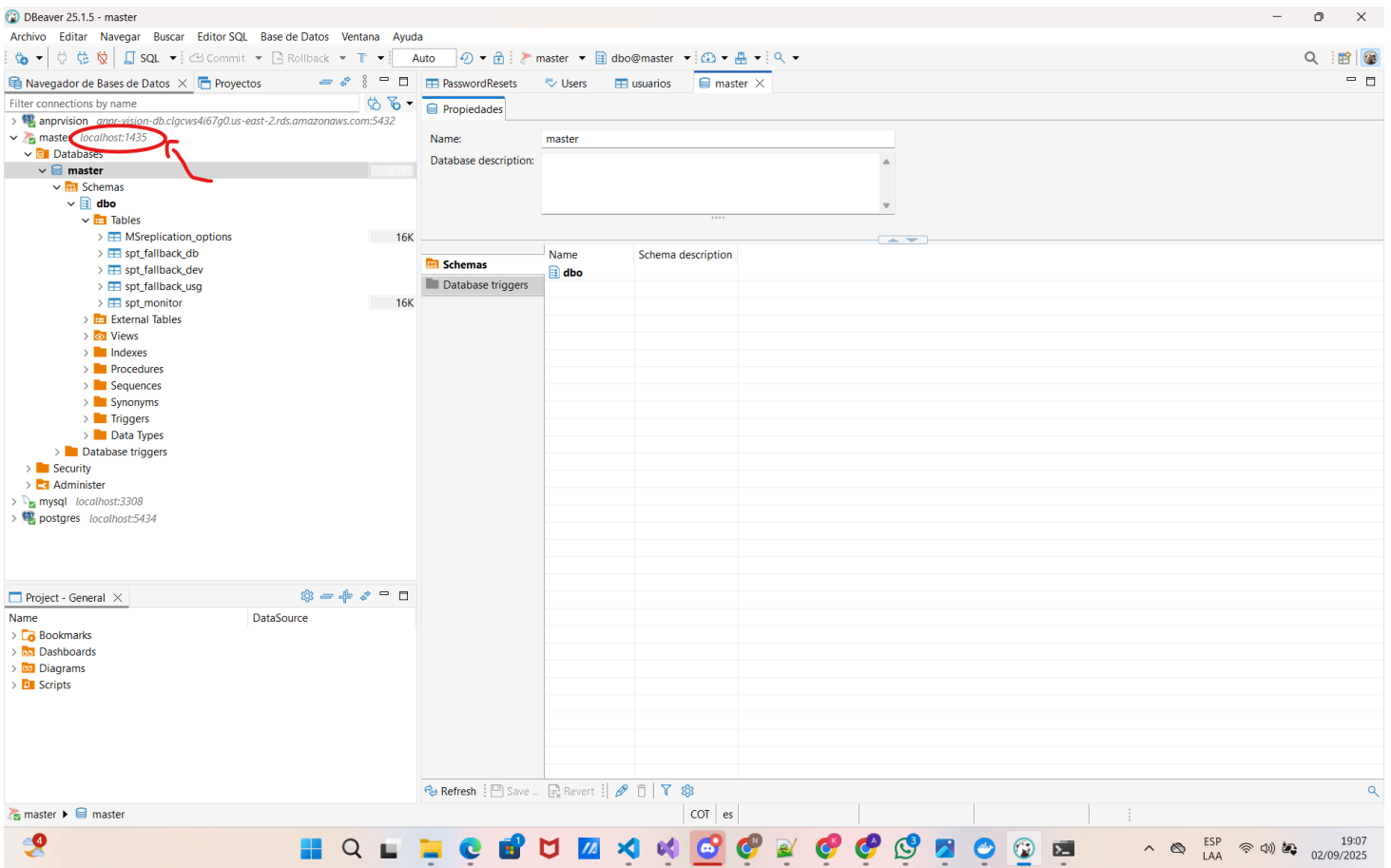
PostgreSQL Database directory appears to contain a database; Skipping initialization

2025-09-02 22:57:46.973 UTC [1] LOG:  starting PostgreSQL 17.6 (Debian 17.6-1.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
2025-09-02 22:57:46.975 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
2025-09-02 22:57:46.975 UTC [1] LOG:  listening on IPv6 address ":::", port 5432
2025-09-02 22:57:46.986 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
2025-09-02 22:57:47.076 UTC [30] LOG:  database system was shut down at 2025-08-25 03:40:49 UTC
2025-09-02 22:57:47.112 UTC [1] LOG:  database system is ready to accept connections
2025-09-02 23:02:47.084 UTC [28] LOG:  checkpoint starting: time
2025-09-02 23:02:47.156 UTC [28] LOG:  checkpoint complete: wrote 3 buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.029 s, sync=0.004 s, total=0.073 s; sync files=2, longest=0.002 s, average=0.002 s; distance=0 kB, estimate=0 kB; lsn=0/1DA23A0, redo lsn=0/1DA2348
```

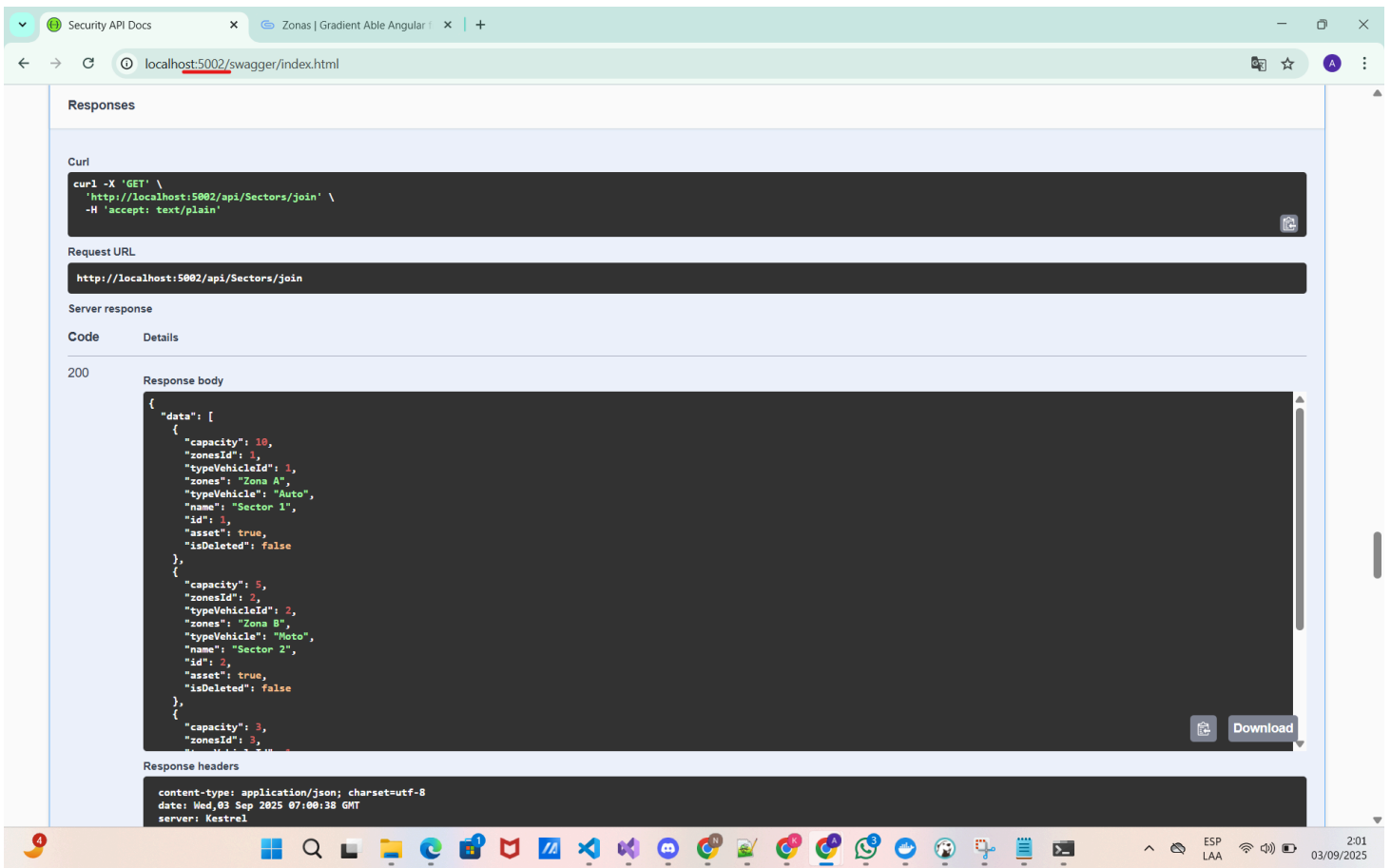
Conexión a SQL Server



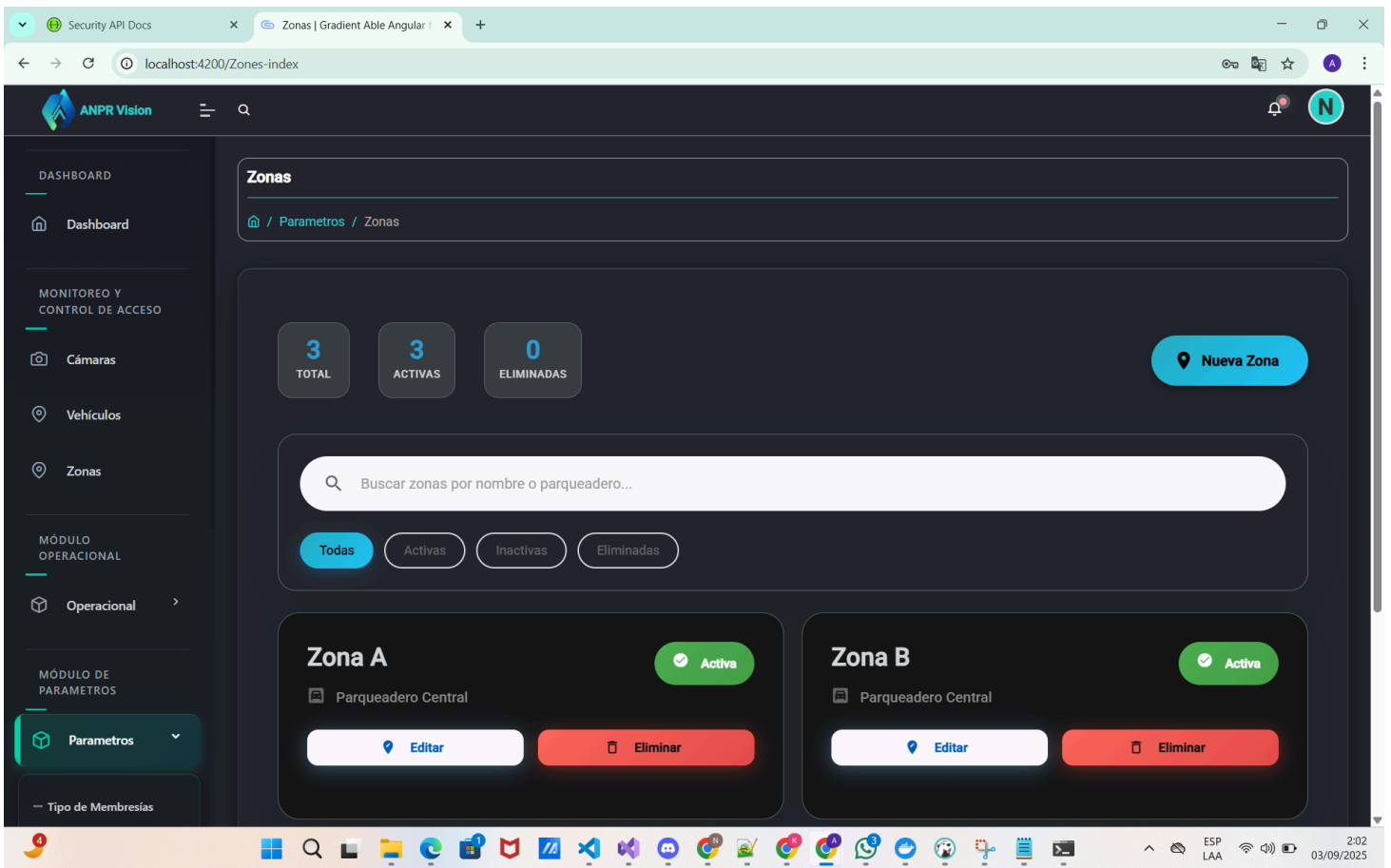


2. Backend y Frontend en Docker apuntando a la base de datos Docker

Vista general: contenedores de apps y motores de BD: La captura de Docker Desktop muestra ejecutándose, al mismo tiempo, los contenedores del backend, del frontend y de los motores de base de datos. Esta vista confirma que todos los servicios están arriba, comparten la misma red de Docker y pueden comunicarse por nombre de servicio.



Frontend operativo y conectado al backend: La siguiente captura muestra el frontend en la pantalla de Zonas. Se visualizan tarjetas como Zona A y Zona B con sus estados, junto con filtros y acciones (Editar, Eliminar). Este resultado evidencia que el frontend está consumiendo la API y renderizando datos reales provenientes de la base de datos del entorno Docker.



Verificación de datos en el cliente SQL: En la parte inferior se ve el cliente de base de datos (DBeaver) conectado al motor en Docker. Se inspeccionan esquemas y tablas del proyecto (por ejemplo, Zonas, Sectors, MembershipTypes, Rates, etc.). Esta comprobación cierra el circuito end-to-end: los datos mostrados en el frontend son los mismos que residen en la BD de Docker y que la API consulta en tiempo real.

DBEAVER 25.1.5 - dbo

ArchivoEditarNavegarBuscarEditor SQLBase de DatosVentanaAyuda

Auto

masterdbo@masterusuariosanprvisionMysqlanprvisionServerdboUsersdbo

Navegador de Bases de DatosProyectos

Filter connections by name

anprvisionanpr-vision-db.cicgw5467g0.us-east-2.rds.amazonaws.com:5432

anprvisionMysqllocalhost:3308

masterlocalhost:1436

Databases

anprvisionServer

Schemas

dbo

Tables

BlackList144K

Camera144K

Clients144K

FormModule216K

Forms72K

MembershipTypes72K

Memberships216K

Modules72K

ParkingCategories72K

Parkings144K

PasswordResets72K

Permissions72K

Persons72K

Rates288K

RatesTypes72K

RegisteredVehicles216K

Rol72K

RolFormPermission288K

RolUser216K

Sectors216K

Slots144K

TypeVehicles72K

Users144K

Vehicles216K

Zones144K

_EFMigrationsHistory72K

External Tables

Views

Indexes

PropiedadesDiagrama

Name:dbo

ID:1

Schema database:anprvisionServer

Schema description:

	Table Name	Type	Create Date	Modify Date	Total bytes	Used bytes	Object description (comment)
Tables	BlackList	USER_TABLE	2025-09-03 05:43:12.397	2025-09-03 05:43:25.147	147.456	32.768	
External Tables	Camera	USER_TABLE	2025-09-03 05:43:06.877	2025-09-03 05:43:26.113	147.456	32.768	
Views	Clients	USER_TABLE	2025-09-03 05:43:54.723	2025-09-03 05:43:54.73	147.456	32.768	
Indexes	FormModule	USER_TABLE	2025-09-03 05:43:54.703	2025-09-03 05:43:54.71	221.184	49.152	
Procedures	Forms	USER_TABLE	2025-09-03 05:43:01.313	2025-09-03 05:43:54.71	73.728	16.384	
Sequences	MembershipTypes	USER_TABLE	2025-09-03 05:43:01.32	2025-09-03 05:43:12.41	73.728	16.384	
Synonyms	Memberships	USER_TABLE	2025-09-03 05:43:12.407	2025-09-03 05:43:26.173	221.184	49.152	
Triggers	Modules	USER_TABLE	2025-09-03 05:43:01.32	2025-09-03 05:43:54.707	73.728	16.384	
Data Types	ParkingCategories	USER_TABLE	2025-09-03 05:43:01.323	2025-09-03 05:43:54.717	73.728	16.384	
DDL	Parkings	USER_TABLE	2025-09-03 05:43:54.713	2025-09-03 05:43:54.727	147.456	32.768	
	PasswordResets	USER_TABLE	2025-09-03 05:43:10.523	2025-09-03 05:43:26.183	0	0	
	Permissions	USER_TABLE	2025-09-03 05:43:01.33	2025-09-03 05:43:06.863	73.728	16.384	
	Persons	USER_TABLE	2025-09-03 05:43:02.51	2025-09-03 05:43:54.733	73.728	16.384	
	Rates	USER_TABLE	2025-09-03 05:43:06.89	2025-09-03 05:43:26.993	294.912	65.536	
	RatesTypes	USER_TABLE	2025-09-03 05:43:02.52	2025-09-03 05:43:06.893	73.728	16.384	
	RegisteredVehicles	USER_TABLE	2025-09-03 05:43:12.417	2025-09-03 05:43:27.01	221.184	49.152	
	Rol	USER_TABLE	2025-09-03 05:43:02.523	2025-09-03 05:43:10.533	73.728	16.384	
	RolFormPermission	USER_TABLE	2025-09-03 05:43:06.86	2025-09-03 05:43:27.05	294.912	65.536	
	RolUser	USER_TABLE	2025-09-03 05:43:10.53	2025-09-03 05:43:27.07	221.184	49.152	
	Sectors	USER_TABLE	2025-09-03 05:43:10.537	2025-09-03 05:43:27.093	221.184	49.152	
	Slots	USER_TABLE	2025-09-03 05:43:12.413	2025-09-03 05:43:28.39	147.456	32.768	
	TypeVehicles	USER_TABLE	2025-09-03 05:43:02.53	2025-09-03 05:43:10.537	73.728	16.384	
	Users	USER_TABLE	2025-09-03 05:43:54.733	2025-09-03 05:43:54.743	147.456	32.768	
	Vehicles	USER_TABLE	2025-09-03 05:43:08.72	2025-09-03 05:43:28.44	221.184	49.152	
	Zones	USER_TABLE	2025-09-03 05:43:08.72	2025-09-03 05:43:28.45	147.456	32.768	
	_EFMigrationsHistory	USER_TABLE	2025-09-03 05:43:01.19	2025-09-03 05:43:01.19	73.728	16.384	

RefreshSave ...Revert

Schema: dbo

COTes

Project - General

NameDataSource

Bookmarks

masteranprvisionServerdbo

2:0403/09/2025